

The workshop has mixed up its labels

Learning intention

Describe burning and rusting as changes that form new materials, and distinguish them from a change of state.

A torn ticket and a rusty bracket both look different.

Did both become a new substance?

Three things we can recover

R1. How could liquid water become ice?

R2. Does dissolved salt stop existing?

R3. Does melting always make a new substance?

I ask two different questions

Can we undo it? Did it make a new substance?

ICE → LIQUID WATER

Cooling can reverse it.
The substance is still water.
Physical change.

PAPER → TORN PAPER

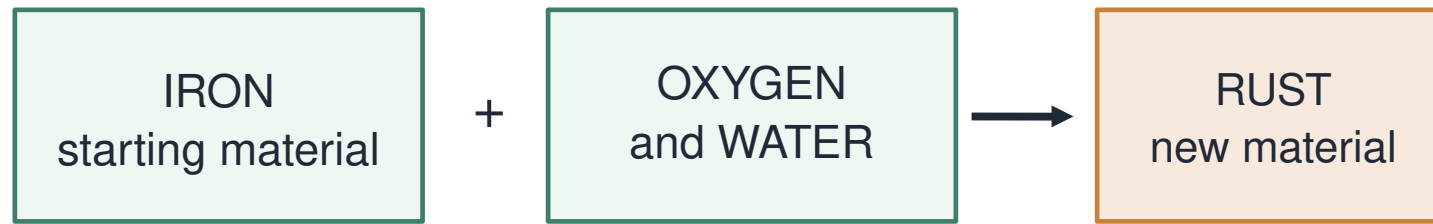
Hard to restore the sheet.
The paper is still paper.
Physical change.

Being hard to undo is not enough evidence
that a new substance formed.

Can a simple process reverse this change? Is the material still the same substance? I use the second question to classify the change.

I explain the rusty bracket

Rusting makes new substances

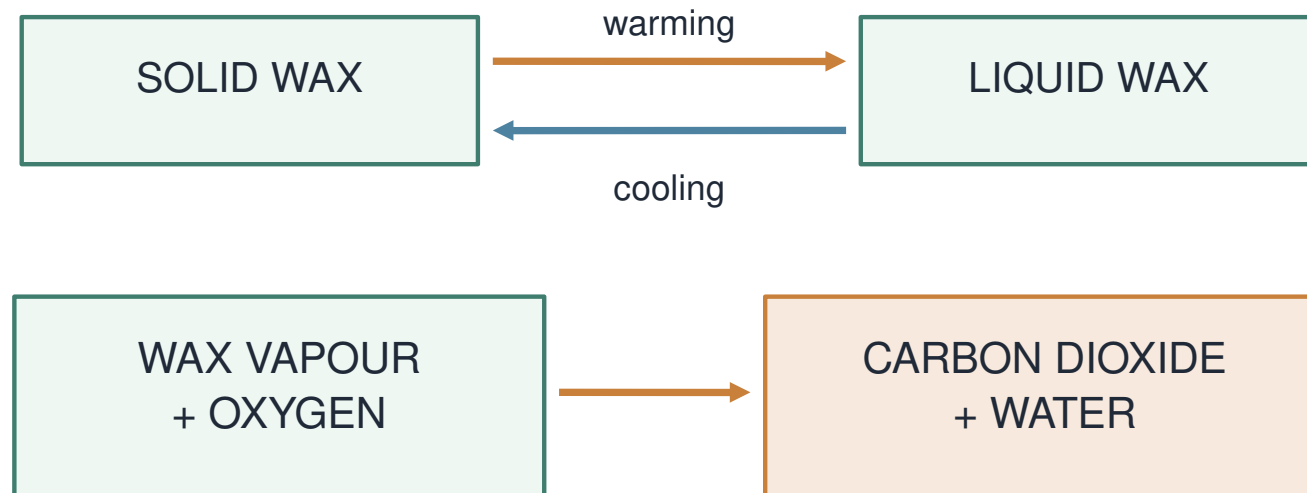


Cooling or wiping rust does not make it iron again.
Word model: not a balanced chemical equation.
Rusting is gradual; animation time is not real time.

Iron reacts with oxygen and water. Rust is a new material. Cooling does not restore the original iron.

I separate two candle processes

Melting is different from burning



Burning forms new substances. This word model omits incomplete-combustion products.

Melting wax changes its state. Burning wax vapour makes new substances. Cooling reverses melting, not burning.

The evidence sorting board

Read cards A–D on shared page 1. W1. Sort by whether new substances formed. Give the evidence, not just a letter.

Case	Key evidence
A: ice	Water can freeze again
B: paper	Same paper, different shape
C: bracket	New rust material identified
D: wood	Ash and gases produced

Repair the workshop rule

W2. Test the rule against torn paper.

Replace it with a better decision question.

Which case supports your new rule?

Two labels for one candle

W3. Label E-A and E-B separately.

Which part can cooling reverse?

Where does the burnt wax material go?

Which is the strongest evidence?

A sample looks different after a change. Which fact most supports a chemical change?

- A** It has a different shape.
- B** Tests identify a new substance.
- C** It is awkward to put back.

Write a workshop diagnosis

Choose one response route.

Classify with evidence and explain rusting or burning.

Make a label a younger visitor can understand.

Use a cause, not only a description

“It turned brown” describes an observation.

What new material formed?

What reacted to produce it?

Check the two-change trap

Have you separated wax melting from burning?

Have you avoided “everything disappears”?

Use one accurate reason per diagnosis.

Audit the diagnosis

Share privately, with a partner, or with the teacher.

Underline the evidence for a new substance.

Correct one misleading rule.

Choose the next exhibit

Should the workshop show rusting or candle changes?

Explain which would clear up a misconception.

We will use one reason to improve the display.

Leave a reliable label

E1. Why is rusting a chemical change?

E2. Why can torn paper still be physical?

E3. What is different about wax burning?